

Solutions – Quiz 3

ECON 320 – Introduction to Mathematical Economics

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Part A – Multiple Choice (1 mark each)

1. (c) $f'(a) = 0$ and $f''(a) < 0$ for a local maximum.
2. (b) $f'(x) = 0$ and $f''(x) > 0$ for a local minimum.
3. (c) $f(x) = x^3 - 6x^2 + 9x$. $f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3)$, so $x = 1, 3$.
 $f''(x) = 6x - 12$; $f''(1) = -6 < 0$ (max), $f''(3) = 6 > 0$ (min). Minimum at $x = 3$.
4. (b) Profit is maximized where marginal *profit* is zero, equivalently $MR = MC$. Given the answers, “zero” is correct.
5. (b) For $y = xe^{-x}$, $y' = e^{-x}(1 - x)$, set to zero $\Rightarrow x = 1$ (maximum).
6. (a) $f(x) = a - bx^2$ with $b > 0$ has $f''(x) = -2b < 0$, strictly concave.
7. (c) The necessary condition for an interior extremum is $f'(x) = 0$.

Part B – Analytical Problems (13 marks)

Question 1 (4 marks)

A firm's profit: $\pi(Q) = 100Q - 4Q^2 - 20$.

(i) FOC.

$$\pi'(Q) = 100 - 8Q.$$

(ii) Maximizer Q^* . Set $\pi'(Q) = 0 \Rightarrow 100 - 8Q = 0 \Rightarrow Q^* = \frac{100}{8} = 12.5$.

(iii) Maximum profit.

$$\pi(12.5) = 100(12.5) - 4(12.5)^2 - 20 = 1250 - 625 - 20 = 605.$$

(iv) **SOC.** $\pi''(Q) = -8 < 0$, so Q^* is a strict maximum.

Interpretation. Output increases profit up to $Q = 12.5$; beyond it, diminishing marginal profit turns total profit down.

Question 2 (4 marks)

Utility: $U(x) = 10x - \frac{1}{2}x^2$.

(i) **FOC.** $U'(x) = 10 - x$.

(ii) **Maximizer** x^* . $10 - x = 0 \Rightarrow x^* = 10$.

(iii) **Value at optimum.** $U(10) = 10 \cdot 10 - \frac{1}{2}(10)^2 = 100 - 50 = 50$.

(iv) **SOC.** $U''(x) = -1 < 0$, so x^* is a maximum.

Interpretation. Marginal utility declines linearly; optimal consumption balances benefit and the rising marginal disutility.

Question 3 (5 marks)

Revenue: $R(P) = 500P e^{-0.1P}$.

(i) **FOC.**

$$R'(P) = 500 [e^{-0.1P} + P(-0.1)e^{-0.1P}] = 500e^{-0.1P}(1 - 0.1P).$$

Set $R'(P) = 0 \Rightarrow 1 - 0.1P = 0 \Rightarrow P^* = 10$.

(ii) **Maximum revenue.**

$$R(10) = 500 \cdot 10 \cdot e^{-1} = \frac{5000}{e} \approx 1839.4.$$

(iii) **SOC.** Differentiate again:

$$R''(P) = 500e^{-0.1P}(-0.1(1 - 0.1P) - 0.1) = 500e^{-0.1P}(-0.2 + 0.01P).$$

At $P = 10$: $R''(10) = 500e^{-1}(-0.1) < 0$, so a strict maximum.

(iv) **Economic reasoning.** At low prices, raising P increases revenue; beyond $P = 10$, the exponential demand response dominates, quantity falls fast enough that total revenue declines.